

Ch8

Supervised learning (classification)

Supervision: The training data (observations, measurements, etc.) are accompanied by labels indicating the class of the observations

- New data is classified based on the training set

Unsupervised learning (clustering)

The class labels of training data is unknown

- Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Prediction Problems: Classification vs. Numeric Prediction

Classification: predicts categorical class labels (discrete or nominal)

classifies data (constructs a model) based on the training set and the values (class labels) in classifying attribute and uses it in classifying new data

Numeric Prediction

models continuous-valued functions, i.e., predicts unknown or missing values

Typical applications

Credit/loan approval, Medical diagnosis ,Fraud detection: if a transaction is fraudulent احتيال ,Web page categorization

Classification—A Two-Step Process

1. بناء النموذج (Model Construction):

Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**

- ▣ The set of tuples used for model construction is **training set**

2. استخدام النموذج (Model Usage):

for classifying future or unknown objects

□ Estimate accuracy of the model

The known label of test sample is compared with the classified result from the model

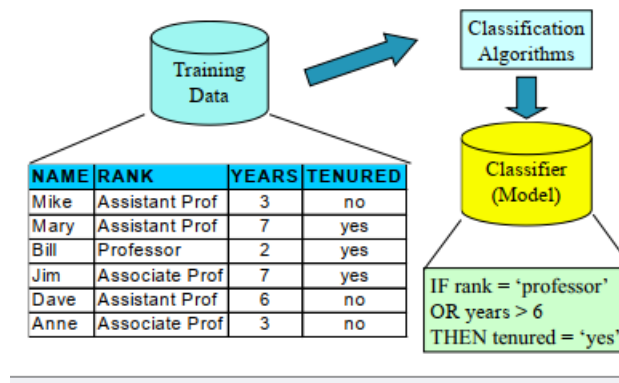
□ Accuracy rate is the percentage of test set samples that are correctly classified by the model

□ **Test set** is independent of training set (otherwise overfitting) **Overfitting** (يعني الجهاز يتعلم (الاختبار تكون مختلفة عن مجموعة التدريب، علشان ما يحصلش (البيانات اللي عنده بالضبط وما يقدرش يعمم على بيانات جديدة).

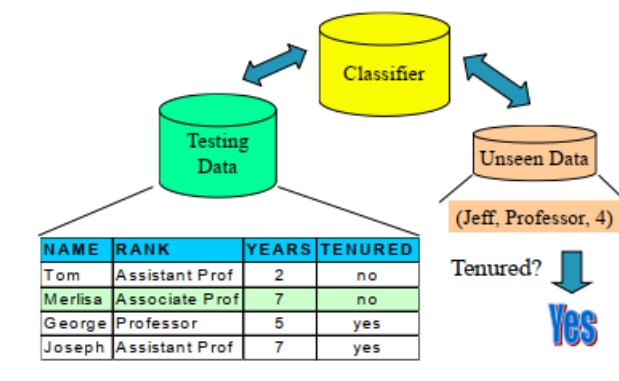
□ If the accuracy is acceptable, use the model to **classify new data**

□ Note: **If the test set is used to select models, it is called validation (test) set**

Process (1): Model Construction



Process (2): Using the Model in Prediction



Decision Tree

Algorithm for Decision Tree Induction

- Basic algorithm (a greedy algorithm)
 - Tree is constructed in a **top-down recursive divide-and-conquer manner**
 - At start, all the training examples are at the root
 - Attributes are **categorical (if continuous-valued, they are discretized in advance)**
 - Examples are partitioned recursively based on selected attributes
 - Test attributes are selected on the basis of a heuristic or statistical measure (e.g., **information gain**)
- Conditions for stopping partitioning
 - All samples for a given node **belong to the same class**
 - There are no remaining attributes for further partitioning – **majority voting** is employed for classifying the leaf
 - There are no samples left

الخوارزمية الأساسية:

- **Greedy Algorithm:**

يعني الخوارزمية بتأخذ قراراتها خطوة بخطوة، وكل مرة تختار التقسيم الأفضل بناءً على الحالة الحالية

- **بداية البناء:**

- في البداية (Root) يتكون في الجذر (Training Examples) كل أمثلة التدريب
- قبل البدء (Categorical) ، بنحولها لقيم فئات (Continuous Values) إذا البيانات كانت قيم مستمرة

- **كيفية التقسيم:**

- بشكل متكرر (Attributes) البيانات بتتقسم بناءً على السمات
- **Information Gain** (السمة التي بتقسم عليها بنختارها بناءً على معيار إحصائي أو مقياس معين (اللي يقيس مدى الفائدة من التقسيم باستخدام السمة دي

متى نوقف التقسيم؟

1. كل الأمثلة في عقدة معينة تنتمي لنفس الفئة

يعني ما فيش حاجة تاني نقسم عليها لأن كل البيانات وصلت لنفس التصنيف

2. ما فيش سمات متبقية للتقسيم

(Majority Voting) لو خلصنا كل السمات وما فيش حاجة نستخدمها للتقسيم، بنستخدم تصويت الأغلبية (Leaf) لتحديد تصنيف العقدة الأخيرة

3. ما فيش أمثلة متبقية

لو كل البيانات تم استخدامها وما فضلش حاجة لتقسيمها

النتيجة:

- شجرة القرار بتكون جاهزة وبتستخدم لتصنيف بيانات جديدة عن طريق متابعة القرارات (التقسيمات) من الجذر للأوراق
- لو البيانات مش منظمة بشكل جيد **Overfitting** الخوارزمية دي فعالة لكنها ممكن تعاني من

Entropy

- هو مقياس لمدى عدم اليقين أو العشوائية في مجموعة من البيانات
- كلما كانت البيانات أكثر عشوائية، كان الإنتروبي أعلى
- على العكس، إذا كانت البيانات منظمة بشكل كامل (لا يوجد عشوائية)، فالإنتروبي يكون منخفض جدًا أو ص

Brief Review of Entropy

■ Entropy (Information Theory)

- A measure of uncertainty associated with a random variable

- Calculation: For a discrete random variable Y taking m distinct values $\{y_1, \dots, y_m\}$,

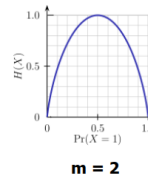
$$H(Y) = -\sum_{i=1}^m p_i \log(p_i), \text{ where } p_i = P(Y = y_i)$$

- Interpretation:

- Higher entropy => higher uncertainty
- Lower entropy => lower uncertainty

■ Conditional Entropy

$$H(Y|X) = \sum_x p(x) H(Y|X = x)$$



Select the attribute with the highest information gain

- Expected information (entropy) needed to classify a tuple in D:

$$Info(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

- Information needed (after using A to split D into v partitions) to classify D:

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$

- Information gained by branching on attribute A

$$Gain(A) = Info(D) - Info_A(D)$$

مساله :

Attribute Selection: Information Gain

- Class P: buys_computer = "yes"

- Class N: buys_computer = "no"

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0)$$

$$+ \frac{5}{14} I(3,2) = 0.694$$

age	p _i	n _i	I(p _i , n _i)
<=30	2	3	0.971
31...40	4	0	0
>40	3	2	0.971

$\frac{5}{14} I(2,3)$ means "age <=30" has 5 out of 14 samples, with 2 yes'es and 3 no's. Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

Similarly,

$$Gain(income) = 0.029$$

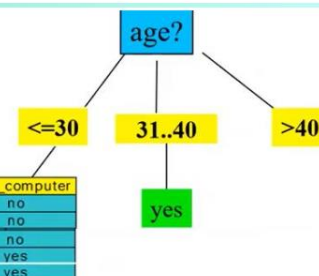
$$Gain(student) = 0.151$$

$$Gain(credit_rating) = 0.048$$

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

For age <= 30 records:

$$Info(D) = I(2,3) = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = 0.97$$



age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

$$Gain_{income}(D) = 0.97 - \left(\frac{2}{5} I(0,2) + \frac{2}{5} I(1,1) + \frac{1}{5} I(1,0)\right) = 0.57$$

$$Gain_{student}(D) = 0.97 - \left(\frac{3}{5} I(0,3) + \frac{2}{5} I(2,0)\right) = 0.97$$

$$Gain_{credit_rating}(D) = 0.97 - \left(\frac{2}{5} I(1,1) + \frac{3}{5} I(1,2)\right) = 0.019$$

We select "student" field to be the split field

1. حساب إجمالي المعلومات (Entropy) للبيانات الأساسية (Info(D)):

المعادلة:

$$Info(D) = -\frac{p}{p+n} \log_2 \frac{p}{p+n} - \frac{n}{p+n} \log_2 \frac{n}{p+n}$$

القيم:

- $p = 9$ (عدد العينات التي بها "yes").
- $n = 5$ (عدد العينات التي بها "no").
- العدد الإجمالي $= 9 + 5 = 14$.

الحساب:

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2 \left(\frac{9}{14}\right) - \frac{5}{14} \log_2 \left(\frac{5}{14}\right) = 0.940$$

لدينا ثلاث مجموعات حسب القيم:

1. $\Rightarrow 30$: 5 عينات (2 "yes" و 3 "no").
2. $31 \dots 40$: 4 عينات (4 "yes" و 0 "no").
3. < 40 : 5 عينات (3 "yes" و 2 "no").

المعادلة لكل تقسيم:

$$I(p_i, n_i) = -\frac{p_i}{p_i + n_i} \log_2 \frac{p_i}{p_i + n_i} - \frac{n_i}{p_i + n_i} \log_2 \frac{n_i}{p_i + n_i}$$

الحساب لكل مجموعة:

1. ≤ 30 :

$$I(2, 3) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

2. $31 \dots 40$:

$$I(4, 0) = -\frac{4}{4} \log_2 \frac{4}{4} - 0 = 0$$

3. > 40 :

$$I(3, 2) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

حساب Info_age:

$$Info_{age}(D) = \frac{5}{14} I(2, 3) + \frac{4}{14} I(4, 0) + \frac{5}{14} I(3, 2)$$

• احسب:

$$Info_{age}(D) = \frac{5}{14} (0.971) + \frac{4}{14} (0) + \frac{5}{14} (0.971)$$

$$Info_{age}(D) = 0.694$$

3. حساب Gain للسمة (Gain(attribute)):

المعادلة:

$$Gain(attribute) = Info(D) - Info_{attribute}(D)$$

للعمر (age):

$$Gain(age) = 0.940 - 0.694 = 0.246$$

4. حساب باقي القيم (income, student, credit_rating):

• يتم حساب $Info_{attribute}$ بنفس الطريقة لكل سمة.

• ثم يتم طرح القيمة من $Info(D)$ لحساب $Gain$.

- Bayesian Classification: Why?+

- ■ A statistical classifier: performs *probabilistic prediction*, i.e., predicts class membership probabilities
- ■ Foundation: Based on Bayes' Theorem.
- ■ Performance: A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- ■ Incremental: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- ■ Standard: Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

■ Total probability Theorem:
$$P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$$

■ Bayes' Theorem:
$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

- Let $\mathbf{X}$ be a data sample ("evidence"): class label is unknown
- Let H be a *hypothesis* that $\mathbf{X}$ belongs to class C
- Classification is to determine $P(H|\mathbf{X})$, (i.e., *posteriori probability*): the probability that the hypothesis holds given the observed data sample $\mathbf{X}$
- $P(H)$ (*prior probability*): the initial probability
 - E.g., $\mathbf{X}$ will buy computer, regardless of age, income, ...
- $P(\mathbf{X})$: probability that sample data is observed
- $P(\mathbf{X}|H)$ (*likelihood*): the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds
 - E.g., Given that $\mathbf{X}$ will buy computer, the prob. that $\mathbf{X}$ is 31..40, medium income

مساله مهمه :

Example 23: a simple binary communication channel carries messages by using only two signals, 0 and 1. for a given binary channel, 40% of the time a 1 is transmitted; the probability that a transmitted 0 is correctly received is 0.90, and the probability that a transmitted 1 is correctly received is 0.95. Determine (a) the probability of a 1 being received, and (b) given a 1 is received, the probability that 1 was transmitted.

$$P(A) = 0.4, \quad P(A^c) = 0.6$$

$$P(B|A) = 0.95, \quad P(B^c|A) = 0.05$$

$$P(B^c|A^c) = 0.90 \quad P(B|A^c) = 0.10$$

$$P(A \cap B) = P(A|B)P(B)$$

(a) We wish to find $P(B)$. Since A and A^c are mutually exclusive and exhaustive:

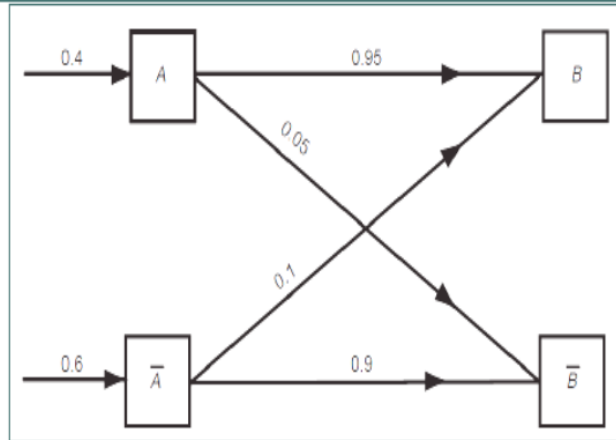
$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) = 0.95(0.4) + 0.1(0.6) = 0.44$$

(b) Using Bayes' theorem

$$1 - 0.44 = 0.56$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{0.95(0.4)}{0.44} = 0.863$$

(c) Find $P(B^c)$.



حل المسائل دي :

<https://drive.google.com/drive/folders/1K2DE9fNdbzRYYV2hsZB5qTG4FilDrW9g?usp=sharing>

Activate

Example 24: For the Trinary communication channel, find the conditional probabilities, $P(B_1 / A_1)$, $P(B_2 / A_1)$, $P(B_2 / A_2)$, $P(B_1 / A_2)$, $P(B_3 / A_2)$, $P(B_2 / A_3)$, $P(B_3 / A_3)$, and all other branch values are illustrated in Figure below.

Solutions:

For three emitting stations $P(B_1) = P(B_2) = P(B_3) = 1/3$;

$P(A_1 / B_1) = 0.8$; $P(A_1 / B_2) = 0.1$;

$P(A_2 / B_1) = 0.2$; $P(A_2 / B_2) = 0.8$; $P(A_2 / B_3) = 0.2$;

$P(A_3 / B_2) = 0.1$; $P(A_3 / B_3) = 0.8$;

$P(A_1) = P(B_1) [P(A_1 / B_1) + P(A_1 / B_2)] = 0.9/3$

$P(A_2) = P(B_2) [P(A_2 / B_1) + P(A_2 / B_2) + P(A_2 / B_3)] = 1.2/3$;

$P(A_3) = P(B_3) [P(A_3 / B_2) + P(A_3 / B_3)] = 0.9/3$

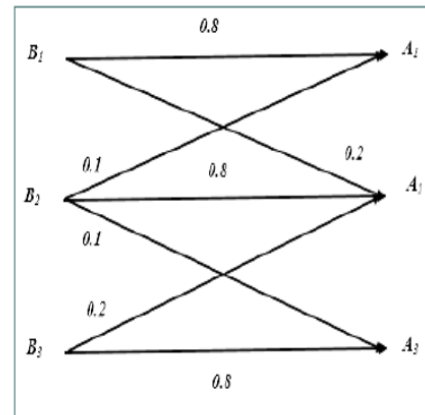
$P(B_1 / A_1) = [P(B_1) P(A_1 / B_1)] / P(A_1) = [0.8/3] / [0.9/3] = 8/9$;

$P(B_2 / A_1) = [P(B_2) P(A_1 / B_2)] / P(A_1) = 1/9$

$P(B_2 / A_2) = [P(B_2) P(A_2 / B_2)] / P(A_2) = 0.8/1.2$

$P(B_1 / A_2) = [P(B_1) P(A_2 / B_1)] / P(A_2) = 1/6$;

$P(B_3 / A_2) = ?$; $P(B_2 / A_3) = ?$; $P(B_3 / A_3) = ?$



Homework 1: An analysis of a long message transmitted in binary digits shown 3000 zeros, and 7000 ones. What is the probability that any one digit in message is a zero?

Ans : $3000/10000 = 0.3$

Home work 2: A box contains 5 white balls, 3 red balls, and 2 black. What is the probability that the 2 balls drawn from the box without replacement with both red ?

Ans : without replacement : يعني لما اخذ الكوره مش هرجعها تاني:

All balls = 10 balls

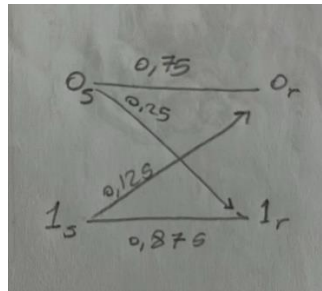
Redballs = 3

عشان هسحب مرتين :

$10C2 = 45$

So $3/45 = 1/15$

Homework 3: In an unsymmetric communication system, a zero or a one is transmitted with $P(X=0) = P_0 = 0.5$, and $P(X=1) = P_1$. Due to noise in the channel, a zero can be received as a one with probability 0.25, and one can be received as a zero with probability 0.125. Calculate all the conditional probability.



1. Probabilities of transmission:

- $P(X = 0) = P_0 = 0.5$
- $P(X = 1) = P_1 = 0.5$

2. Conditional probabilities (errors in communication):

- $P(Y = 1 | X = 0) = 0.25$ (a zero is received as one).
- $P(Y = 0 | X = 1) = 0.125$ (a one is received as zero).

3. Correct reception probabilities:

- $P(Y = 0 | X = 0) = 1 - P(Y = 1 | X = 0) = 1 - 0.25 = 0.75$.
- $P(Y = 1 | X = 1) = 1 - P(Y = 0 | X = 1) = 1 - 0.125 = 0.875$.

1. Calculate $P(Y = 0)$:

The total probability of receiving a zero ($Y = 0$) can be found using the law of total probability:

$$P(Y = 0) = P(Y = 0 | X = 0) \cdot P(X = 0) + P(Y = 0 | X = 1) \cdot P(X = 1)$$

Substitute the values:

$$P(Y = 0) = (0.75 \cdot 0.5) + (0.125 \cdot 0.5)$$

$$P(Y = 0) = 0.375 + 0.0625 = 0.4375$$

2. Calculate $P(Y = 1)$:

Similarly, the total probability of receiving a one ($Y = 1$):

$$P(Y = 1) = P(Y = 1 | X = 0) \cdot P(X = 0) + P(Y = 1 | X = 1) \cdot P(X = 1)$$

Posterior Probabilities (Bayes' Rule):

Using Bayes' rule, we calculate the conditional probabilities $P(X = 0 | Y = 0)$, $P(X = 1 | Y = 0)$, $P(X = 0 | Y = 1)$, and $P(X = 1 | Y = 1)$.

3. Calculate $P(X = 0 | Y = 0)$:

$$P(X = 0 | Y = 0) = \frac{P(Y = 0 | X = 0) \cdot P(X = 0)}{P(Y = 0)}$$

Substitute the values:

$$P(X = 0 | Y = 0) = \frac{0.75 \cdot 0.5}{0.4375} = \frac{0.375}{0.4375} \approx 0.8571$$

4. Calculate $P(X = 1 | Y = 0)$:

$$P(X = 1 | Y = 0) = \frac{P(Y = 0 | X = 1) \cdot P(X = 1)}{P(Y = 0)}$$

Substitute the values:

$$P(X = 1 | Y = 0) = \frac{0.125 \cdot 0.5}{0.4375} = \frac{0.0625}{0.4375} \approx 0.1429$$

Summary of Results:

1. $P(Y = 0) = 0.4375$
2. $P(Y = 1) = 0.5625$
3. $P(X = 0 | Y = 0) \approx 0.8571$
4. $P(X = 1 | Y = 0) \approx 0.1429$
5. $P(X = 0 | Y = 1) \approx 0.2222$
6. $P(X = 1 | Y = 1) \approx 0.7778$

Homework 4: Consider a digital signal is transmitted through three stages in sequence. Each stage performs independently. Suppose the digit one is transmitted, and because of presence of noise in circuits there is a finite probability that the input digits may be inverted by each stage. A list of the following probabilities of inversion for each of the three stages respectively are :
1st stage: $P(1 \rightarrow 0) = 1/4$, $P(0 \rightarrow 1) = 1/2$; 2nd stage : $P(1 \rightarrow 0) = 1/2$, $P(0 \rightarrow 1) = 1/4$
3rd stage : $P(1 \rightarrow 0) = 1/4$, $P(0 \rightarrow 1) = 1/4$. Calculate the probability that a one is received at the end of the third stage, if the input digit to the first stage is a one ? What is the frequency of the number of inversion ?

Prediction Based on Bayes' Theorem:

- Given training data $\mathbf{X}$, *posteriori probability* of a *hypothesis* H , $P(H|\mathbf{X})$, follows the Bayes' theorem

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

posteriori = likelihood x prior/evidence

- الاحتمال اللاحق (Posterior): هو الاحتمال الذي نريد حسابه، وهو احتمال أن تنتمي البيانات (X) إلى فئة معينة (C) بعد أخذ البيانات التدريبية بعين الاعتبار.
- احتمال الاحداث (Likelihood): هو احتمال الحصول على البيانات (X) إذا كانت تنتمي بالفعل إلى الفئة (C).
- الاحتمال السابق (Prior): هو احتمال أن تنتمي البيانات إلى الفئة (C) قبل رؤية أي بيانات.
- الدليل (Evidence): هو احتمال الحصول على البيانات (X) بغض النظر عن الفئة.

- Predicts $\mathbf{X}$ belongs to C_i iff the probability $P(C_i|\mathbf{X})$ is the highest among all the $P(C_k|\mathbf{X})$ for all the k classes

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

- $P(C_i): P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$$

- Compute $P(X|C_i)$ for each class

$$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$$

$$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$$

$$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$$

$$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$$

- **$X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$**

$$P(X|C_i): P(X|\text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$$

$$P(X|\text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$$

$$P(X|C_i) \cdot P(C_i): P(X|\text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$$

$$P(X|\text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = 0.007$$

Therefore, X belongs to class ("buys computer = yes")

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Num of yes